

Xing Yao

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Education

Vanderbilt University

Doctor of Philosophy, Computer Science

Nashville, TN

08/2021 - 05/2026

Duke University

Master of Science, Biomedical Engineering

Durham, NC

08/2019 - 05/2021

Xi'an Jiaotong University (XJTU)

Bachelor, Biomedical Engineering

Xi'an, China

09/2015 - 06/2019

Working Experiences

1) Development of Video Segmentation Foundation Model

05/2025 – 09/2025

GE HealthCare, AI/ML Ph.D. Internship

- Led development of a **200M-parameter** SAM2-based video segmentation foundation model trained on **200K+ image-mask pairs** across **multi-center datasets**.
- Developed an **agent-driven automated data curation pipeline** for large-scale video datasets, performing statistical profiling (video duration/object counts/object scale distribution) and anomaly detection to remove low-quality samples.
- Built **config-driven experimentation workflows** that automatically schedule training jobs on available GPUs, orchestrate hyperparameter sweeps and ablation studies, and integrate experiment monitoring and logging for scalable and reproducible model development.
- Developed a **multi-prompt inference evaluation pipeline** to systematically benchmark model performance across diverse prompt formulations/prompt ensemble sizes, dataset scales, and target object sizes.
- Enhanced model efficiency and robustness through architecture scaling (**18M-220M parameters**), initialization strategies, spatio-temporal data augmentation, and staged training strategies (**mid-training, joint training, fine-tuning**).
- Delivered a scalable training and inference system supporting internal clinical QA workflows. ([Link](#))

2) Conditional Latent Diffusion for Robust Video Temporal Super-Resolution

05/2024 - 08/2024

GE HealthCare, AI/ML Ph.D. Internship

- Led the development of the first conditional latent diffusion framework for ultrasound video frame interpolation, enabling high-fidelity temporal super-resolution under heavy noise and complex tissue motion.
- Designed a **multi-stage embedding-level conditioning architecture** to preserve conditioning signals across deep diffusion networks, mitigating information attenuation and improving temporal consistency in generated frames.
- Achieved **>70%** improvement in perceptual and temporal metrics (LPIPS, FVD) over prior motion-estimation-based methods on large-motion ultrasound videos.
- Integrated model into GEHC's internal workflows to improve stability of downstream segmentation systems. ([Link](#)).

Selected Projects

MedICL Lab, Vanderbilt University

Nashville, TN

1) Multi-Modal Geometric-Semantic feature Fusion for 3D Ultrasound Registration

- Developed a **training-free 3D ultrasound registration framework** that fuses **vision foundation model (DINOv3)-derived semantic features** with **point-cloud-based geometric features** to handle large inter-volume motion.
- Designed **coarse-to-fine** matching strategy achieving **near real-time performance (2.6s/pair)**, and **coarse-only real-time (0.1s/pair, 2.5GB)** performance with **17%** improvement in registration accuracy compared to SOTA methods.
- Enabled efficient and robust 3D registration under large relative motion and real-world high noise situation ([Link](#))

2) Latent Diffusion-Based 3D Outpainting for Efficient 3D Ultrasound Stitching

- Developed a **self-supervised 3D latent diffusion-based outpainting framework**, enabling **3D outpainting** without paired training data, overcoming the sector-shaped field-of-view limitation inherited from US imaging.
- Enabled inference-time 3D completion on large missing regions. Reduced inference time from **40min to 59s per batch** and memory from **759MB to 21MB** compared to SOTA methods.
- Mitigated the local minima introduced by FOV effect and significantly (**p<0.05**) improved registration accuracy by **80%**.
- The work was nominated for MIDL2025 best oral paper (top 9). ([Link](#))

3) ControlNet-based Synthetic Data Generation for Self-Supervised Training

- Developed a **patch-conditioned ControlNet diffusion model** capable of generating realistic ultrasound panoramas from minimal conditioning signals (**~10% image patch**).

- Designed a **controllable synthetic data engine** that generates paired images with self-defined spatial transformations (translation, rotation, scaling, shearing).
- Simulated real-world acquisition artifacts (speckle noise, shadowing, probe motion) to produce high-fidelity training data.
- Enabled scalable downstream registration training without paired real-world data, achieving significant ($p < 0.05$) performance improvements (**>15%**) over traditional and learning-based baselines. ([Link](#))

4) Robust Foundation Model Adaptation with Uncertainty-Guided Calibration

- Designed an **uncertainty-guided** segmentation system built on top of Segment Anything (SAM), improving robustness to **noisy medical images** and **unstable prompts**.
- Introduced **uncertainty calibration** and **false positive/negative control**, resulting in **+10%** segmentation accuracy (Dice) improvement and **-52%** in structural error (ASSD) compared to the original SAM.
- Significantly ($p < 0.05$) improved robustness to coarse or imprecise bounding-box prompts in real-world usage.
- Extended the system from **2D to 3D**, enabling accurate 3D volume segmentation from a single 2D bounding box. ([Link](#))

Computational Optics Lab, Duke University

Durham, NC

1) Large Field-of-View Microscopy System with ML-Based Imaging and Classification

- Led the development of a **multi-lens overlapping microscopy system** integrated with a **DL-based classification pipeline**, enabling a significantly expanded effective field of view at low hardware cost.
- Achieved **7× faster** imaging throughput and cell detection compared to traditional single-lens microscopy setups. ([Link](#))

Technical Skills

- **Tools:** PyTorch, CUDA, Docker, Git, Hugging Face, scikit-learn, pandas, MatLab, MONAI
- **Core AI/ML:** Generative AI, Foundation Models, Diffusion Models, Vision-Language Models, Contrastive Learning, Self-Supervised Learning, Transformer, Segmentation, Classification, Registration

Academic Service

Conference Reviewer: CVPR, MICCAI, MIDL, Medical Image Analysis, Biomedical Signal Processing and Control

Teaching Assistant: Reinforcement Learning, SQL Database Management (Vanderbilt University, 08/2021 - 12/2022)

Guest lecture Speaker: Open-Source Programming in Medical Image Processing (Vanderbilt University, 2021 – 2026)

Publications

First-author publications

- [1] **Yao X**, DiSanto N, Yu R, Wang J, Lu D, Arenas GA, Oguz B, Pouch AM, Schwartz N, Byram B, Oguz I. Vision Foundation Model-Empowered Multi-Feature-Guided Ultrasound Stitching Framework. In Medical Imaging with Deep Learning 2026.
- [2] **Yao X**, Yu R, Lu D, DiSanto N, Aghdam EK, Oguine K, Arenas G, Oguz B, Pouch A, Schwartz N, Byram BC. From geometry to intensity: a coarse-to-fine pipeline for unsupervised 3D ultrasound stitching. In Medical Imaging 2026: Image Processing 2026 Apr 3 (Vol. 13925, pp. 655-660). SPIE.
- [3] **Yao X**, Gangopadhyay A, Chang HM, Soni R. Towards Better Ultrasound Video Segmentation Foundation Model: An Empirical study on SAM2 Finetuning from Data Perspective. arXiv preprint arXiv:2511.05731. 2025 Nov 7.
- [4] **Yao X**, Yu R, DiSanto N, Aghdam EK, Oguine KJ, Lu D, Lou A, Wang J, Hu D, Arenas GA, Oguz B. LOTUS: Latent Outpainting Diffusion Model for Three-Dimensional Ultrasound Stitching. In Medical Imaging with Deep Learning 2025.
- [5] **Yao X**, Yu R, Hu D, Yang H, Lou A, Wang J, Lu D, Arenas G, Oguz B, Pouch A, Schwartz N. SynStitch: A Self-Supervised Learning Network for Ultrasound Image Stitching Using Synthetic Training Pairs and Indirect Supervision. In 2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI) 2025 Apr 14 (pp. 1-5). IEEE.
- [6] **Yao X**, Chang HM, Gangopadhyay A, Addala AV, Soni R. Self-supervised ultrasound video frame interpolation based on conditional latent diffusion model. In Medical Imaging 2025: Image Processing 2025 Apr 11 (Vol. 13406, pp. 697-704). SPIE.
- [7] **Yao X**, Liu H, Hu D, Lu D, Lou A, Li H, Deng R, Arenas G, Oguz B, Schwartz N, Byram BC. FNPC-SAM: uncertainty-guided false negative/positive control for SAM on noisy medical images. In Medical Imaging 2024: Image Processing 2024 Apr 2 (Vol. 12926, p. 1292602). SPIE.
- [8] **Yao X**, Lou A, Li H, Hu D, Lu D, Liu H, Wang J, Stoebner Z, Johnson H, Long JD, Paulsen JS. Novel application of the attention mechanism on medical image harmonization. In Medical Imaging 2023: Image Processing 2023 Apr 3 (Vol. 12464, pp. 192-202). SPIE.
- [9] **Yao X**, Pathak V, Xi H, Chaware A, Cooke C, Kim K, Xu S, Li Y, Dunn T, Chandra Konda P, Zhou KC. Increasing a microscope's effective field of view via overlapped imaging and machine learning. Optics Express. 2022 Jan 6;30(2):1745-61.

Co-authored publications

- [1] Yu R, **Yao X**, Huo Y, Chang C. Enhancing skin lesion classification using segmentation-based global-local feature fusion. In Medical Imaging 2025: Image Processing 2025 Apr 11 (Vol. 13406, pp. 756-762). SPIE.

- [2] Li H, Oguz B, Arenas G, **Yao X**, Wang J, Lu D, Pouch A, Byram B, Schwartz N, Oguz I. PRISM Lite: A lightweight model for interactive 3D placenta segmentation in ultrasound. In *Medical Imaging 2025: Image Processing* 2025 Apr 11 (Vol. 13406, pp. 65-70). SPIE.
- [3] Deng R, Yang Y, Pisapia DJ, Liechty BL, Zhu J, Xiong J, Guo J, Lu Z, Wang J, **Yao X**, Yu R. CASC-AI: Consensus-aware self-corrective learning for cell segmentation with Noisy labels. In *Medical Imaging with Deep Learning* 2025.
- [4] Wang J, Li H, Hu D, Xu R, **Yao X**, Tao YK, Oguz I. Retinal IPA: Iterative Key Points Alignment for Multimodal Retinal Imaging. In *International Workshop on Medical Optical Imaging and Virtual Microscopy Image Analysis* 2024 Oct 10 (pp. 119-129). Cham: Springer Nature Switzerland.
- [5] Hu D, Li H, Liu H, Wang J, **Yao X**, Lu D, Oguz I. Adaptdiff: cross-modality domain adaptation via weak conditional semantic diffusion for retinal vessel segmentation. In *International Workshop on Simulation and Synthesis in Medical Imaging* 2024 Oct 6 (pp. 13-23). Cham: Springer Nature Switzerland.
- [6] Li H, Oguz B, Arenas G, **Yao X**, Wang J, Pouch A, Byram B, Schwartz N, Oguz I. Interactive segmentation model for placenta segmentation from 3d ultrasound images. In *International Workshop on Advances in Simplifying Medical Ultrasound* 2024 Oct 5 (pp. 132-142). Cham: Springer Nature Switzerland.
- [7] Li H, Liu H, Hu D, **Yao X**, Wang J, Oguz I. Cats v2: hybrid encoders for robust medical segmentation. In *Medical imaging 2024: image processing* 2024 Apr 2 (Vol. 12926, pp. 130-137). SPIE.
- [8] Arenas G, Pouch A, Oguz I, Oguz B, Byram B, **Yao X**, Schwartz N. 448 Deformable Medial Modeling to Generate Novel Shape Features of the Placenta Using Automated versus Manual Segmentations. *Journal of Clinical and Translational Science*. 2024 Apr;8(s1):132-3.
- [9] Lu D, Wu Y, Acar A, **Yao X**, Wu JY, Kavoussi N, Oguz I. ASSIST-U: A system for segmentation and image style transfer for ureteroscopy. *Healthcare Technology Letters*. 2024 Apr;11(2-3):40-7.
- [10] Lu D, Wu Y, **Yao X**, Kavoussi NL, Oguz I. AVA: automated viewability analysis for ureteroscopic intrarenal surgery. In *Medical Imaging 2024: Image-Guided Procedures, Robotic Interventions, and Modeling* 2024 Mar 29 (Vol. 12928, pp. 113-118). SPIE.
- [11] Lou A, Li Y, **Yao X**, Zhang Y, Noble J. Samsnerf: Segment anything model (sam) guides dynamic surgical scene reconstruction by neural radiance field (nerf). In *Medical Imaging 2024: Image-Guided Procedures, Robotic Interventions, and Modeling* 2024 Mar 29 (Vol. 12928, pp. 19-23). SPIE.
- [12] Hu D, Li H, Liu H, **Yao X**, Wang J, Oguz I. MAP: Domain Generalization via Meta-Learning on Anatomy-Consistent Pseudo-Modalities. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* 2023 Oct 8 (pp. 182-192). Cham: Springer Nature Switzerland.
- [13] Liu H, Li H, **Yao X**, Fan Y, Hu D, Dawant BM, Nath V, Xu Z, Oguz I. Colossal: A benchmark for cold-start active learning for 3d medical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* 2023 Oct 1 (pp. 25-34). Cham: Springer Nature Switzerland.
- [14] Hu D, Li H, Liu H, **Yao X**, Wang J, Oguz I. Vesselmorph: domain-generalized retinal vessel segmentation via shape-aware representation. *arXiv preprint arXiv:2307.00240*. 2023 Jul 1.
- [15] Zia A, Bhattacharyya K, Liu X, Berniker M, Wang Z, Nespolo R, Kondo S, Kasai S, Hirasawa K, Liu B, Austin D, **Yao X**. Surgical tool classification and localization: results and methods from the MICCAI 2022 SurgToolLoc challenge. *arXiv preprint arXiv:2305.07152*. 2023 May 11.
- [16] Lou A, Tawfik K, **Yao X**, Liu Z, Noble J. Min-max similarity: A contrastive semi-supervised deep learning network for surgical tools segmentation. *IEEE Transactions on Medical Imaging*. 2023 Apr 10;42(10):2832-41.
- [17] Harfouche M, Kim K, Zhou KC, Konda PC, Sharma S, Thomson EE, Cooke C, Xu S, Kreiss L, Chaware A, Yang X, **Yao X**. Imaging across multiple spatial scales with the multi-camera array microscope. *Optica*. 2023 Apr 6;10(4):471-80.
- [18] Hu D, **Yao X**, Wang J, Tao YK, Oguz I. Deep angiogram: trivializing retinal vessel segmentation. In *Medical Imaging 2023: Image Processing* 2023 Apr 3 (Vol. 12464, pp. 669-673). SPIE.
- [19] Lou A, **Yao X**, Liu Z, Han J, Noble J. Self-supervised surgical instrument 3D reconstruction from a single camera image. In *Medical Imaging 2023: Image-Guided Procedures, Robotic Interventions, and Modeling* 2023 Apr 3 (Vol. 12466, pp. 102-107). SPIE.
- [20] Cooke CL, Kim K, Xu S, Chaware A, **Yao X**, Yang X, Neff J, Pittman P, McCall C, Glass C, Jiang XS. A multiple instance learning approach for detecting covid-19 in peripheral blood smears. *PLOS Digital Health*. 2022 Aug 19;1(8):e0000078.
- [21] Li H, Lu D, **Yao X**, Kavoussi N, Oguz I. Endo-SemiS: Towards Robust Semi-supervised Image Segmentation for Endoscopic Video. In *Medical Imaging with Deep Learning*.
- [22] Zhu J, Tawil C, **Yao X**, Oguz B, Arenas G, Oguz I, Schwartz N, Byram B. Adaptive Spectral Estimation Improves Preeclampsia Prediction when using Short Pulsed Wave Doppler Acquisition Duration. In *2025 IEEE International Ultrasonics Symposium (IUS)* 2025 Sep 15 (pp. 1-4). IEEE.
- [23] Wang J, Li H, **Yao X**, Toubasi A, Vinarsky T, Gheen C, Derwenskus J, Jin C, Dortch R, Xu J, Bagnato F. DEMIST: Decoupled Multi-stream latent diffusion for Quantitative Myelin Map Synthesis. *arXiv preprint arXiv:2511.12396*. 2025 Nov 16.